Appendix A

A.1 Applications of Tangible Landscape

In this section we briefly describe additional applications of Tangible Landscape that we, our students, and our colleagues have developed. For each project we explain the aim, used methods and studied processes.

Modeling Avalanches in High Tatras

Eva Stopkova modeled snow avalanches in the High Tatras mountain range in Slovakia (Stopková 2007, 2008). Avalanche potential is closely related to landscape topography so Stopkova built a physical model of her study area in High Tatras in order to visualize the spatial pattern of topographic parameters and landforms (Fig. A.1). She was able to project the results of avalanche danger assessment over the model, but she was not able usefully compute the assessment using the modified and scanned physical model. The scanned models were not accurate enough for practical applications due to their small scale and the steepness of their slopes. Modeling avalanches with Tangible Landscape would require a larger physical model at a larger map scale. CNC routing can be used to precisely and efficiently fabricate the size of model needed for such an application.

Visualizing the Evolution of Oregon Inlet

Liliana Velasquez Montoya created a projection augmented sand model of Oregon Inlet where she was modeling how storms drive the morphological evolution of barrier islands. To visualize and present the results of her research Montoya

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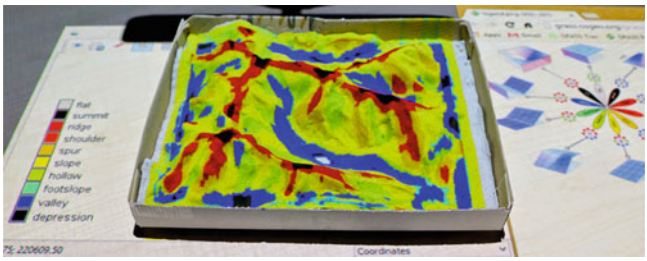

Fig. A.1 A sand model of High Tatras in Slovakia augmented with a projection of the landforms computed by the module *r.geomorphon*; a legend and a diagram of the landform types are projected on the table beside the model

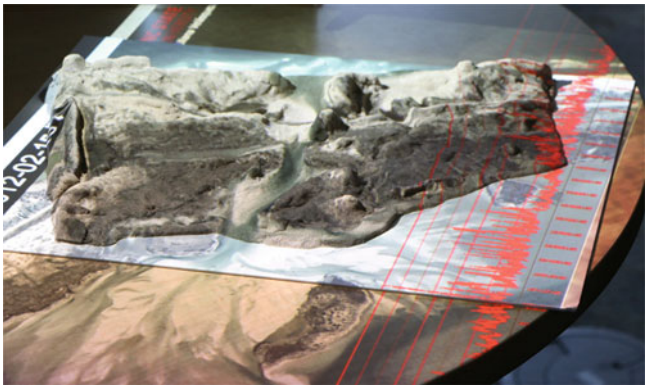

Fig. A.2 A sand model of Oregon Inlet with a projected video showing its formation and evolution along with a graph of wave height changing over time

projected a video depicting the evolution of the inlet in response to storm events onto a sand model. She built the sand model to physically represent the present 3D form of the inlet. Since the sand model was not dynamic it only matched the end of animation. This was a useful mode of representation as the model acted as a datum that helped us compare the present and past morphology of the inlet. Since the small sand model was only large enough to show the most active part of the inlet she also projected the surrounding coastal landscape (Fig. A.2).

Designing Disaster Relief Housing for Rodanthe

Logan Free designed disaster relief shelters for an emergency ferry terminal at Rodanthe on the Outer Banks of North Carolina. The ferry terminal is used for evacuation and relief when there are disasters like hurricanes or bridge failures. Free was tasked with designing temporary housing structures and a community service

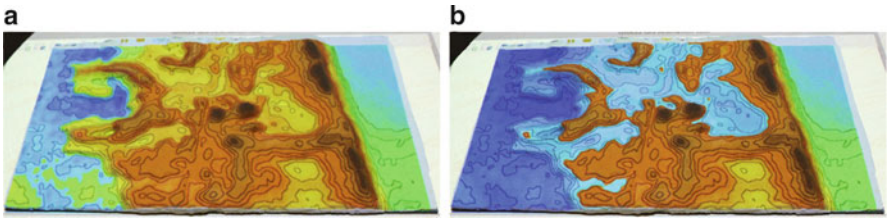

Fig. A.3 Simulating storm surge at the Rodanthe emergency ferry terminal on the Outer Banks of North Carolina: (**a**) the mean high water level on the sound side of the island and (**b**) the approximate water level during high storm surge

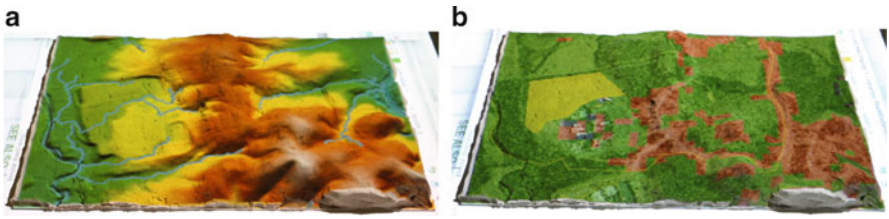

Fig. A.4 Landscape change modeling for a watershed in Charlotte, North Carolina: (**a**) a sand model of the terrain at the Charlotte study area augmented with a projected hillshade to enhance perception of relief; (**b**) using a mouse and the digitizer tool to draw a polygon to designate a development area

center for disaster recovery at the ferry terminal. To identify vulnerable hotspots and select safe sites for the shelters he built a sand model of the site and used Tangible Landscape with the module *r.lake* to simulate the potential extent of storm surge flooding (Fig. A.3).

Simulating Landscape Change in Charlotte

Douglas A. Shoemaker created a model of a small watershed in Charlotte, North Carolina to test how the FUTure Urban-Regional Environment Simulation (FUTURES) framework could be integrated with Tangible Landscape. FUTURES uses a stochastic patch growing algorithm to simulate urban growth and land use change (Meentemeyer et al. 2013). It has been integrated with GRASS GIS as the add-on module *r.futures* (Petrasova et al. 2015). Shoemaker used the raster digitizer tool in the GRASS GIS GUI to designate a site for development and change the land cover. He also sculpted the sand to grade the topography for the new development. Then he ran FUTURES to explore how the new development would influence longterm landscape change in the watershed-scale study area (Fig. A.4).

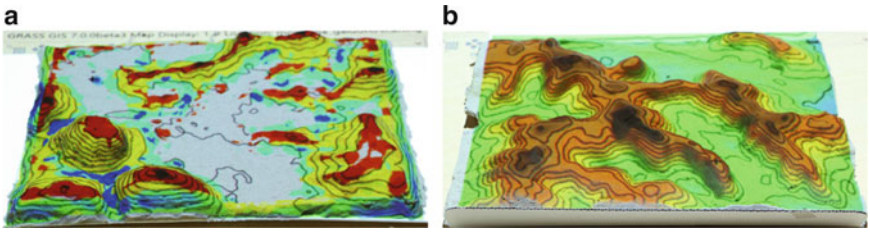

Fig. A.5 Reconstructing a paleolake and a terrain model used for cell tower planning. (**a**) A reconstruction of a paleolake in the Galuut Valley, Mongolia showing the landforms computed by *r.geomorphon*. (**b**) A physical model used to site cell towers in Athens county, Ohio

Reconstructing a Paleolake in Mongolia

Gantulga Bayasgalan used a sand model to reconstruct a paleolake in the Galuut Valley in central southwest Mongolia. He modeled alternative reconstructions of the landscape in sand and used Tangible Landscape to compute the flow accumulation pattern, geomorphological features, and several other characteristics for each landscape configuration (Fig. A.5a).

Cell Tower Planning in Athens County

Jesse Boyd used Tangible Landscape to site cell towers in Athens county, Ohio. To find good sites for cell towers, Boyd tested different spatial configurations by placing markers representing cell towers on a physical model of the landscape (Fig. A.5b). Since topography blocks cellular signals he used viewshed analysis to estimate cellular coverage. He experimented with different cell tower heights by changing the observer elevation parameter of the module *r.viewshed*.

Monitoring Coastal Erosion

Tristan Dyer used terrestrial lidar data to monitor coastal erosion on a beach near the US Army Corps of Engineers' Field Research Facility in Duck on the Outer Banks of North Carolina. Coastal erosion can be visually and tactilely explored by building a sand model for each lidar survey. Dyer scanned a transect of the beach with a terrestrial lidar system, interpolated the lidar point cloud as a DEM, and then built a sand model with the aid of Tangible Landscape to quickly prototype and visualize the landform (Fig. A.6a). A sequence of these models would show the morphological evolution of the beach.

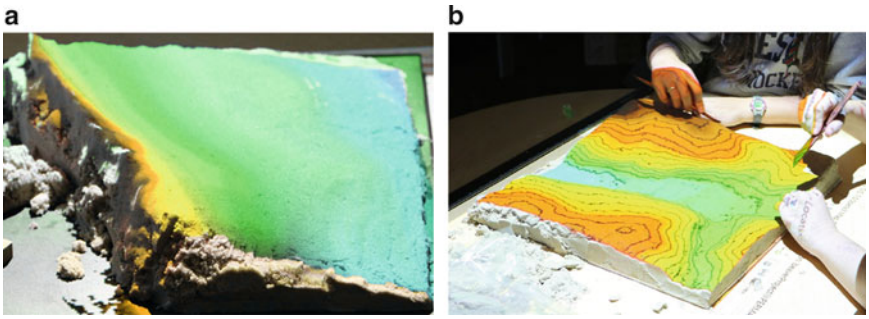

Fig. A.6 Sand models used to study beaches and streams. (**a**) A sand model of a beach in Duck, North Carolina used to study beach erosion. (**b**) A sand model of a valley in North Carolina Piedmont used to study how beaver dams influence stream flow and morphology

Exploring Impacts of a Beaver Dam

Karl W. Wegmann and his students used Tangible Landscape to study how beaver dams affect streams in North Carolina. They used Tangible Landscape to model the morphology and flow of a stream with and without beaver dams. First they built a sand model of a segment of the existing stream with beaver dams and studied its topographical parameters and hydrological processes. Then they sculpted the sand model and computed different water flow scenarios to study how the stream could have been without beaver dams (Fig. A.6b).

In this application we merged the scanned elevation model with a DEM of the surrounding landscape to put the modeled area into its broader context and account for water flowing from the entire contributing watershed. The water flow simulation was then run over the entire area so that the water from the contributing watersheds would fill the stream.

Modeling the Potential Impacts of a Coal Ash Pond Spill

Matthew Horvath (2014) studied the potential impacts of spills from the Cape Fear Plant coal ash ponds near Moncure, North Carolina. Horvath built a sand model for a study area near the Cape Fear river and used Tangible Landscape with the add-on module *r.damflood* to simulate a breach in a coal ash pond (Fig. A.7). Artificial landforms kept the simulated spread of contaminated surface water from reaching the nearby Cape Fear river.

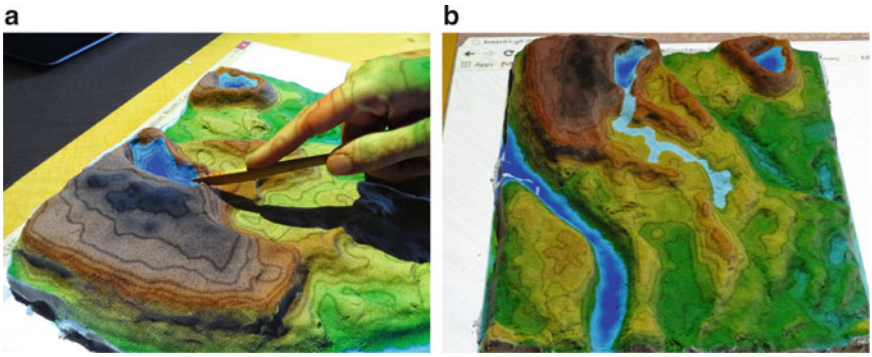

Fig. A.7 Making a breach in a coal ash pond (**a**) and simulating the spread of contaminated water (**b**)

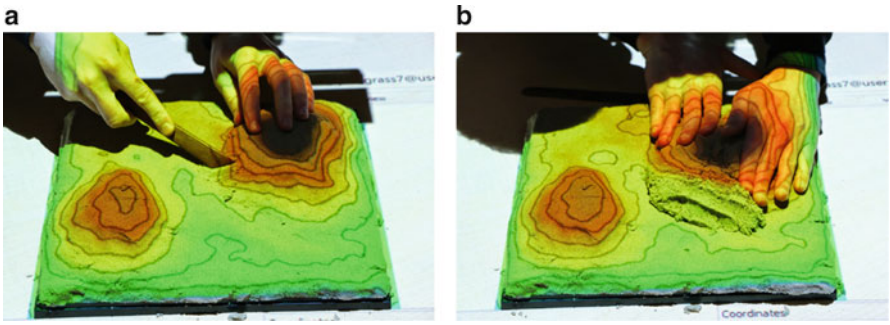

Fig. A.8 Iteratively sculpting a landform in sand to generate time-series data: (**a**) the initial state followed by (**b**) changing the position and shape of the landform

Testing a Landform Migration Algorithm

Petras et al. (2015) used Tangible Landscape to test the behavior of a newly developed algorithm for describing landform migration. The algorithm maps the horizontal migration of complex landforms as gradient fields based on the analysis of contour evolution. The gradient field is a vector field representing the movement and deformation of contours. It can be used to quantify the rate and direction of landform migration at any point in space and time. Data for testing algorithms is usually computer generated. In this case, however we wanted to create simplified, yet realistic abstractions of landforms that would allow us to clearly model and visualize landform migration. It was easier to make these abstract landforms by hand than on the computer. To create test data in a controlled environment we built sand models by hand to represent a moving landform. Using Tangible Landscape we sculpted and scanned a model of the initial terrain and then sculpted and scanned a model of the terrain after it had migrated to create an elevation time-series (Fig. A.8).

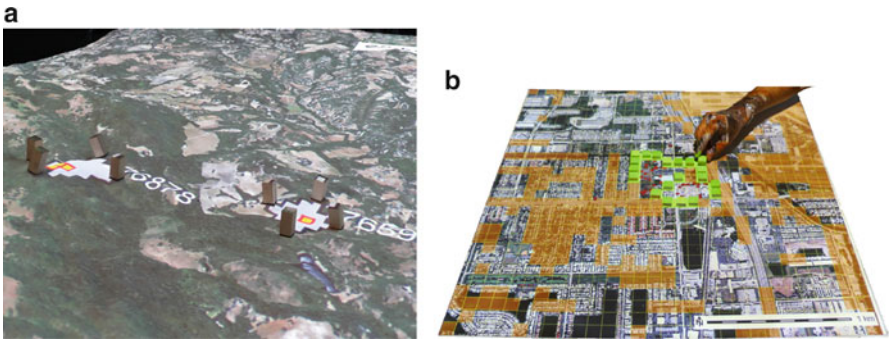

Fig. A.9 Using object recognition to digitize polygons and pixels representing treatment areas in order to computationally steer (**a**) Sudden Oak Death in Sonoma Valley, California, and (**b**) termites spread simulation in Fort Lauderdale, Florida

Managing the Spread of Sudden Oak Death in Sonoma Valley

In collaboration with Francesco Tonini and Douglas A. Shoemaker we used Tangible Landscape to test the effectiveness of different treatment scenarios for controlling the spread of Sudden Oak Death in Sonoma Valley, California. We used a stochastic model to simulate the spread of Sudden Oak Death and Tangible Landscape to visualize the spread and designate new treatment areas. We CNC routed a large (1 m × 1 m) MDF model of the terrain and coated it with magnetic paint. To make markers that would hold in place on the complex topography we attached adhesive magnetic strips to the base of basswood markers. We used the markers to collaboratively draw polygons representing treatment areas with Tangible Landscape (Fig. A.9a). The polygons reduced the host species of the disease on aligned pixels thus influencing the next run of the simulation. Each treatment area was automatically assigned a label with its area. The sum of all of the treatment areas was dynamically displayed on a simple dashboard. The disease spread simulation was written in the R (R Core Team 2013) statistical programing language and environment.

Managing the Spread of Termites in Fort Lauderdale

In collaboration with Francesco Tonini we used Tangible Landscape as a spatial decision support tool for collaboratively managing the spread of termites in Dania Beach, Fort Lauderdale, Florida. We used pixel-based object recognition to interact with a stochastic simulation of the spread of termites. Placing a marker on a flat surface with a projected grid changed a value on the corresponding pixel in the simulation input—a raster index of termite habitat (Fig. A.9b).

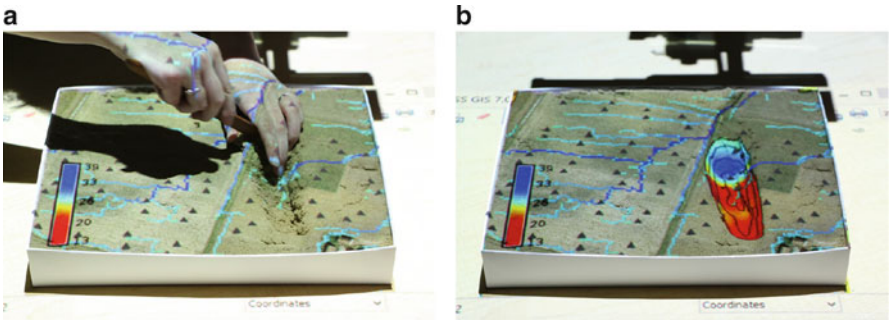

Fig. A.10 Exploring subsurface moisture by removing sand and projecting the cross-section of a scanned surface with the 3D raster of moisture

Tangible Exploration of Subsurface Data

Using Tangible Landscape we explored the 3D distribution of the percentage of subsurface moisture (Petrasova et al. 2014) measured in Kinston, North Carolina and represented as a GRASS GIS 3D raster (GRASS Development Team 2015). By digging into the sand model with our hands we could explore underground as if we were at an excavation site. As we dug into the sand the cross-section of the scanned surface with the 3D raster representing soil properties was projected in real-time (Fig. A.10). We also projected additional GIS layers (such as an orthoimagery, elevation contours, and flow accumulation) over the model to provide spatial context and gain further insights into soil moisture distribution and its relation to the landscape.

Additionally, in combination with marker detection from Sect. 6.2.2 we can create and visualize vertical profiles or soil core samples. This method is an intuitive and natural way of exploring subsurface data and it represents an alternative to more abstract 3D computer visualization tools.

A.2 Data Sources

Throughout the book we use various data sources. Here is a list of the most useful data sources for the study areas presented in the book.

Sample Data for This Book

NCSU OSGeo Research and Education Laboratory

http://geospatial.ncsu.edu/osgeorel/data.html

US Lidar Data

United States Interagency Elevation Inventory

http://coast.noaa.gov/inventory/

Earth Explorer

http://earthexplorer.usgs.gov/

Digital Coast

http://coast.noaa.gov/dataviewer/

Open Topography

http://www.opentopography.org/

US Digital Elevation Models

National Elevation Dataset on the National Map Viewer

http://viewer.nationalmap.gov/viewer/

US Orthoimagery

USGS EROS Orthoimagery web mapping service

http://raster.nationalmap.gov/arcgis/services/Orthoimagery/USGS_EROS_Ortho_1Foot/ImageServer/WMSServer?

US Soil Surveys

USDA Web Soil Survey

http://websoilsurvey.sc.egov.usda.gov

US Fire Modeling Data

LANDFIRE

http://www.landfire.gov/

Global Datasets

GRASS GIS community list of global datasets
http://grasswiki.osgeo.org/wiki/Global_datasets

NC Climate Data

State Climate Office of North Carolina
http://www.nc-climate.ncsu.edu/

A.3 Starting with GRASS GIS

To run examples from this book we provide a dataset which can be downloaded from the NCSU OSGeo Research and Education Laboratory website.[1] Once the downloaded dataset is unzipped we create a directory `grassdata` in our home directory and place the dataset there. Then we start GRASS GIS and a start up dialog in Fig. A.11 appears on screen. In the upper text field we specify the path to our `grassdata` directory. Then we should be able to see Location `nc_spm_tl` containing several Mapsets. We select mapset `practice1` and press the Start button.

After opening the Layer Manager and Map Display windows we display the raster map `elevation` as shown in Fig. A.12. To execute commands from this book presented in boxes with green background we paste the commands into the *Command console* tab as shown in Fig. A.13a and press Enter. The Python code in boxes with blue background can be executed in the *Python shell* tab (Fig. A.13b) or in any Python shell running in a GRASS GIS session.

More instructions how to use GRASS GIS GUI are available on the official GRASS GIS introduction page[2] and in the GRASS GIS GUI manual.[3]

[1] http://geospatial.ncsu.edu/osgeorel/data.html

[2] http://grass.osgeo.org/grass70/manuals/helptext.html

[3] http://grass.osgeo.org/grass70/manuals/wxGUI.html

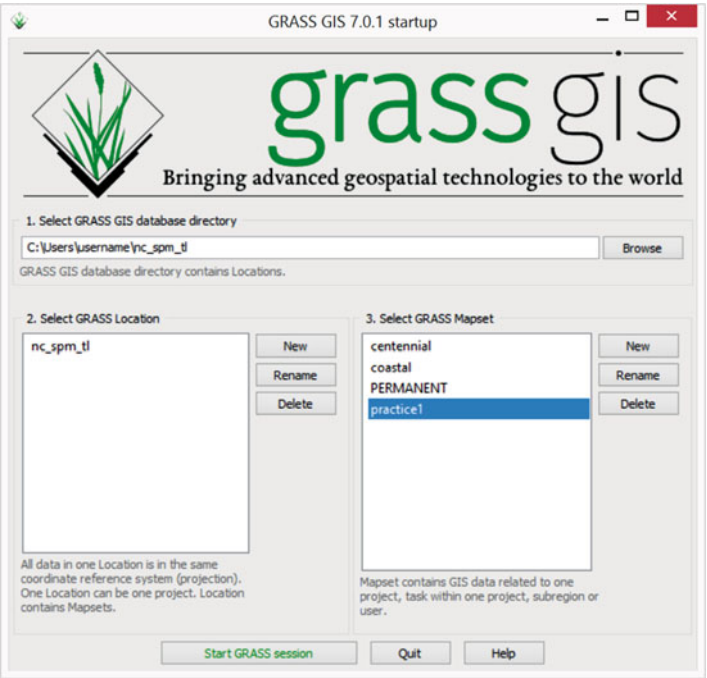

Fig. A.11 Starting GRASS GIS

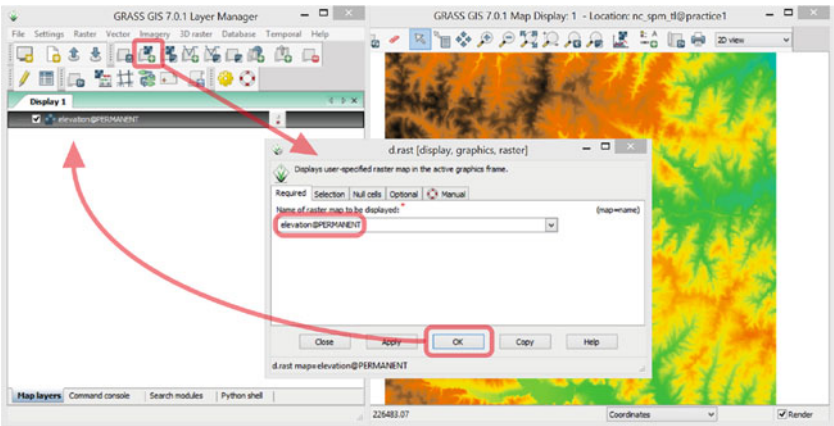

Fig. A.12 Display raster map elevation: click on *Add raster map layer* button in toolbar, select `elevation` from drop-down list, and press OK

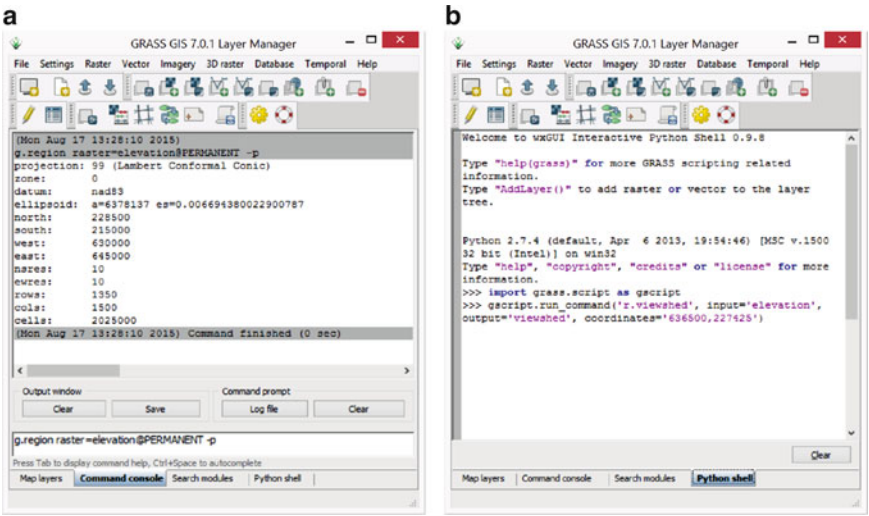

Fig. A.13 GRASS GIS modules can be run from (**a**) the *Command console* tab or (**b**) Python shell using Python syntax

References

GRASS Development Team. 3D raster data in GRASS GIS [online]. (2015) [Accessed 2015-08-13]. http://grass.osgeo.org/grass70/manuals/raster3dintro.html.

Horvath, M. (2014). Spill impacts from coal ash pond using GRASS GIS. Upublished poster.

Meentemeyer, R. K., Tang, W., Dorning, M. A., Vogler, J. B., Cunniffe, N. J., and Shoemaker, D. A. (2013). FUTURES: multilevel simulations of emerging urban-rural landscape structure using a stochastic patch-growing algorithm. *Annals of the Association of American Geographers*, 103(4):785–807.

Petras, V., Mitasova, H., and Petrasova, A. (2015). *Mapping gradient fields of landform migration*, pages 173–176. Bogucki Wydawnictwo Naukowe, Adam Mickiewicz University in Poznań - Institute of Geoecology and Geoinformation, Poznań, Poland.

Petrasova, A., Harmon, B., Mitasova, H., and White, J. (2014). Tangible Exploration of Subsurface Data. Poster presented at 2014 Fall Meeting, AGU, San Francisco, Calif., 15–19 Dec.

Petrasova, A., Petras, V., Shoemaker, D. A., Dorning, M. A., and K. Meentemeyer, R. (2015). The integration of land change modeling framework FUTURES into GRASS GIS 7. In *Geomatics Workbooks n 12 – "FOSS4G Europe Como 2015"*.

R Core Team (2013). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. http://www.R-project.org/.

Stopková, E. (2007). Modelovanie oblastí vzniku lavín s využitím GIS. In *Sborník studentské konference GISáček*.

Stopková, E. (2008). Predikcia lavínového nebezpečenstva s využitím GIS. Bachelor's thesis, Slovak University of Technology in Bratislava.

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